

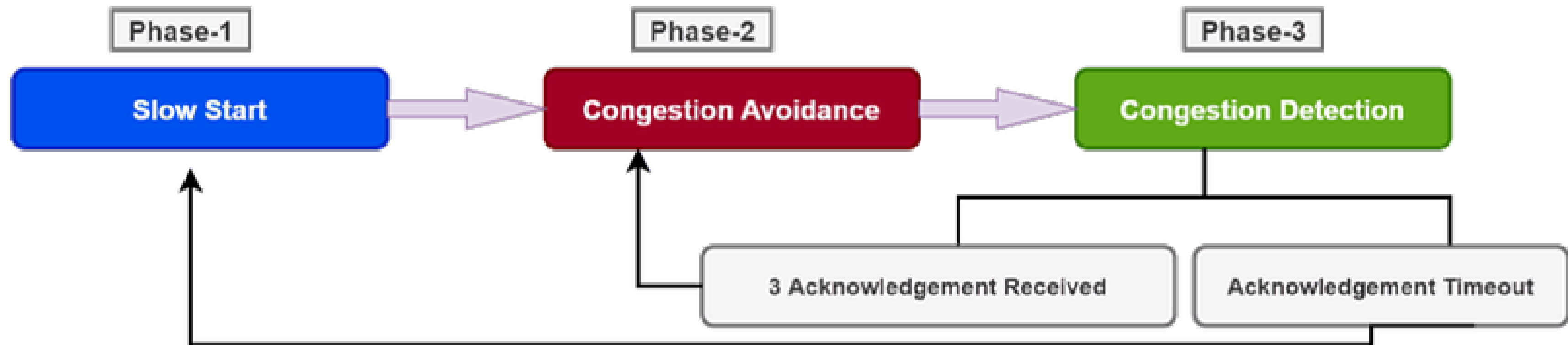


Cloud Load Balancing: ALB vs. NLB, Elastic Load Balancing, and High Availability

Join the lecture online on *your* dashboard

Recap

TCP Congestion Control

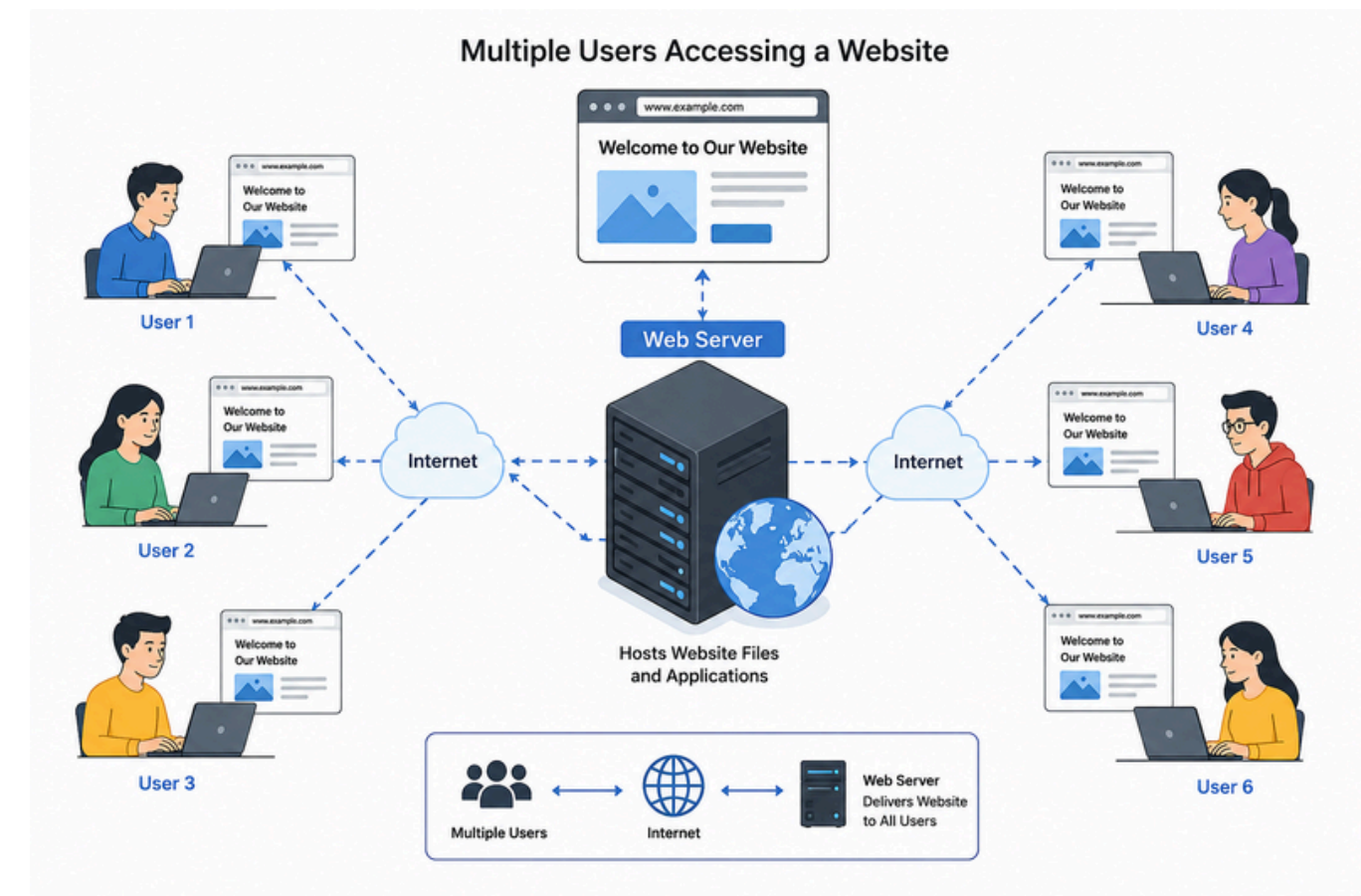


- TCP is a polite driver.
- Its mission: Get your data to its destination as fast as possible, but without causing a traffic jam for everyone else.

Can Our Website Survive the Internet

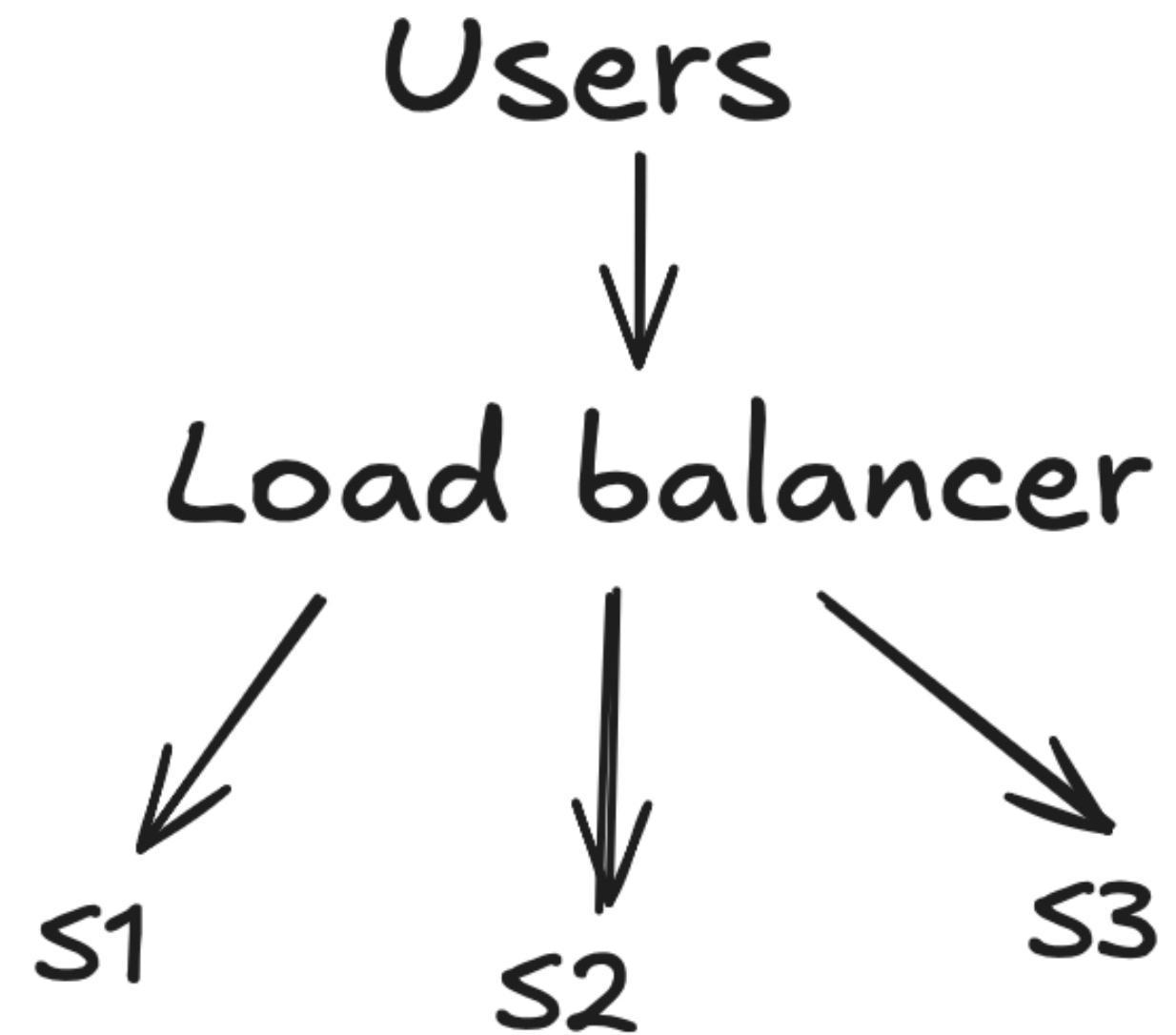
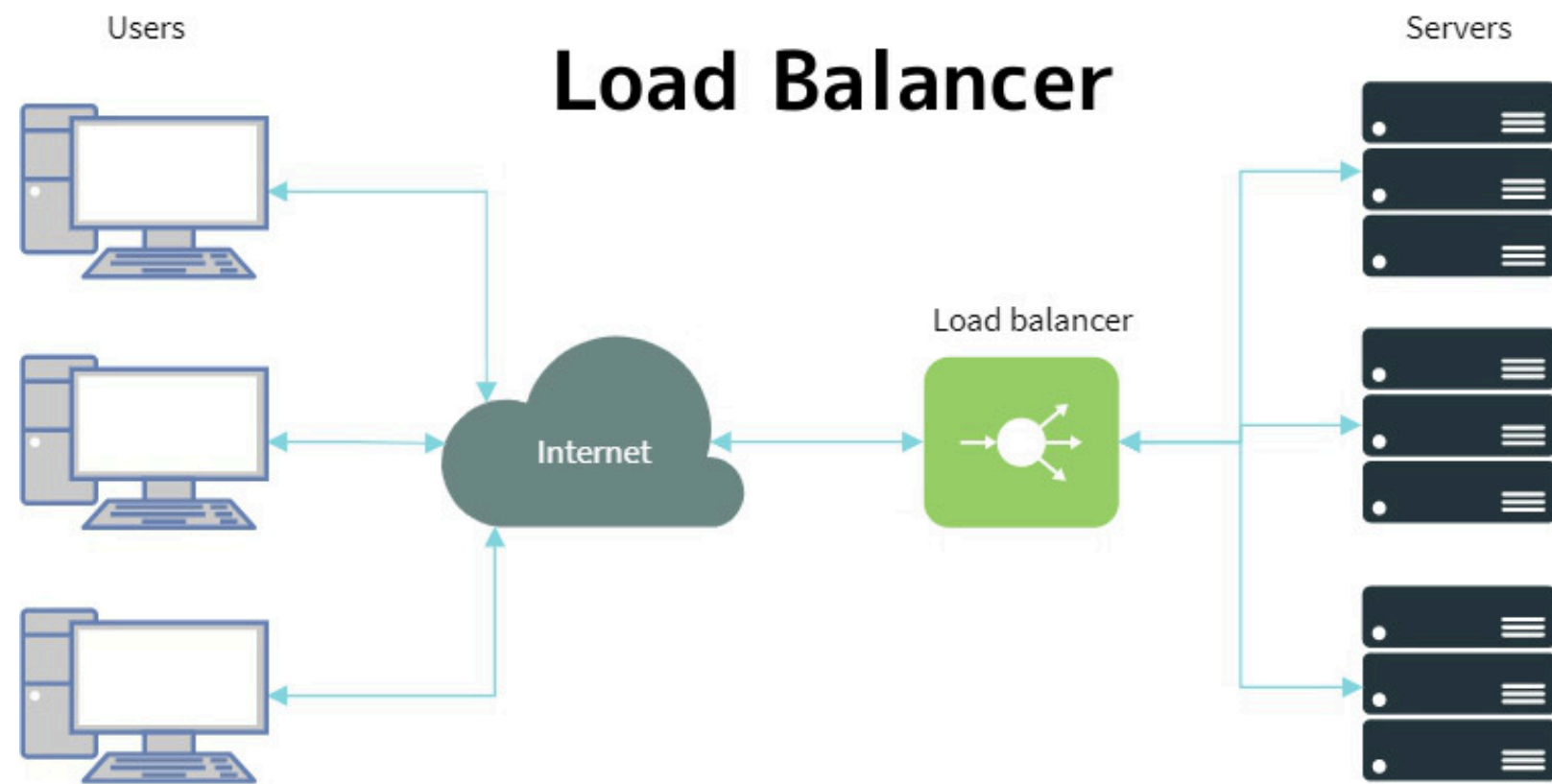
Imagine:

- 10,000 users are trying to access your website.
- One server suddenly crashes.



Load Balancer

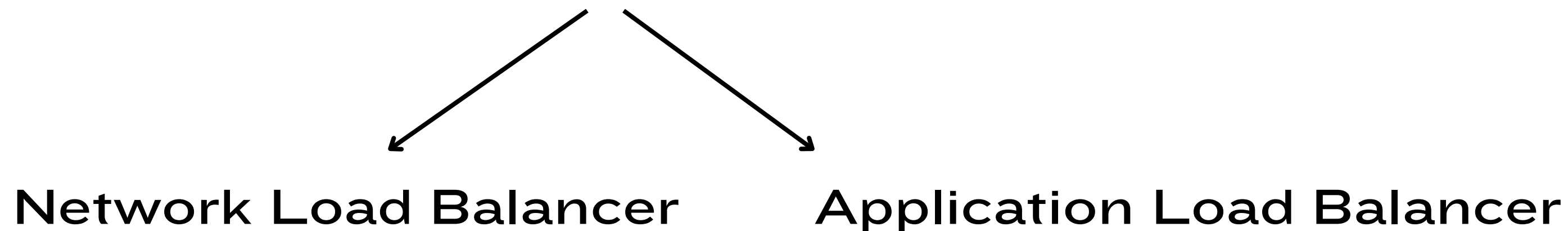
A load balancer distributes incoming network traffic across multiple servers to ensure high availability, reliability, scalability, and optimal application performance.



Load Balancer

How load balancer makes its decision

On the basis of two approaches:



Network Load Balancer(NLB)

Recap: The Transport Tuple

A TCP/UDP connection is identified by a 4-tuple: Source IP, Source Port, Destination IP, Destination Port.

- NLB = Network Load Balancer operating at Transport Layer.
- It sees TCP/UDP packets — not HTTP requests, not URLs, not headers.
- **Analogy:** A postal sorter who reads only the ZIP code, not the letter inside.

Network Load Balancer(NLB)

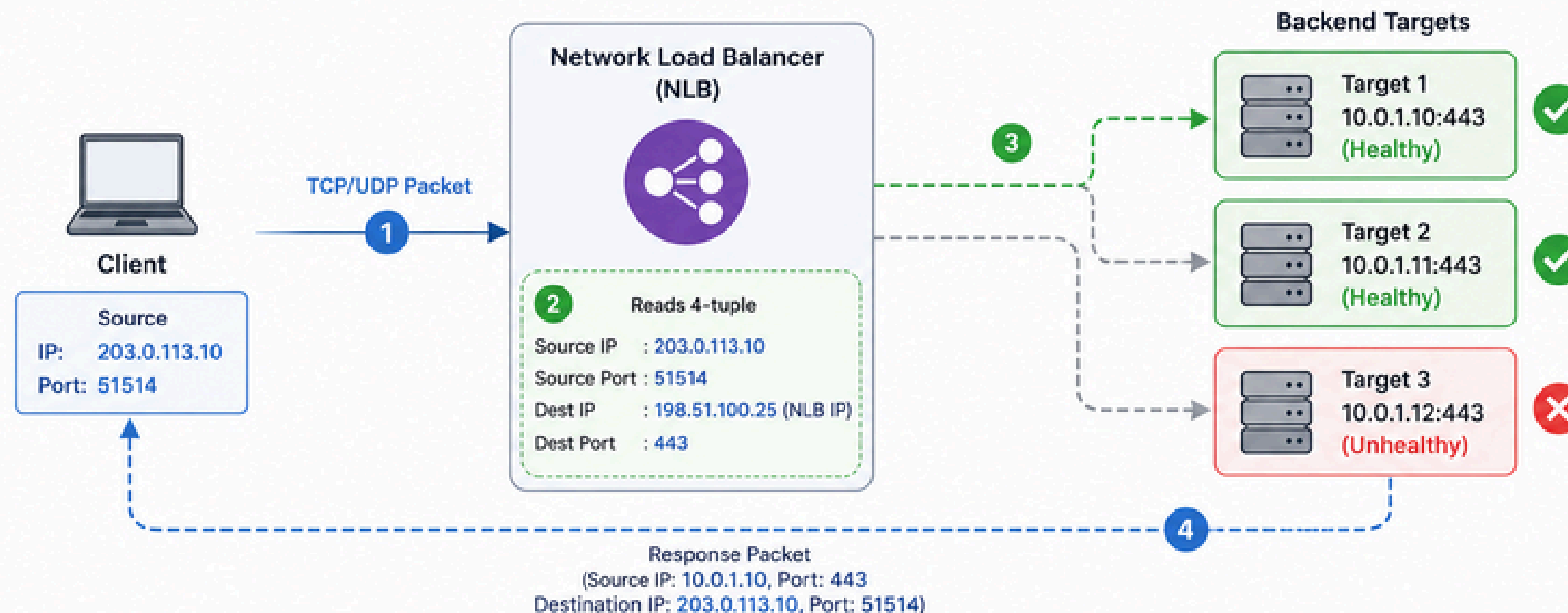
How NLB Forwards Traffic (The Actual Mechanism)

1 Client connects →
NLB receives the
TCP/UDP packet

2 NLB reads the
4-tuple (src IP, src port,
dst IP, dst port)

3 Forwards the connection
to a healthy
backend target

4 The client's original
source IP is preserved
end-to-end



Key Points

- NLB operates at Layer 4 (TCP/UDP)
- It makes forwarding decisions based on the 4-tuple
- Client's original source IP is preserved to the backend

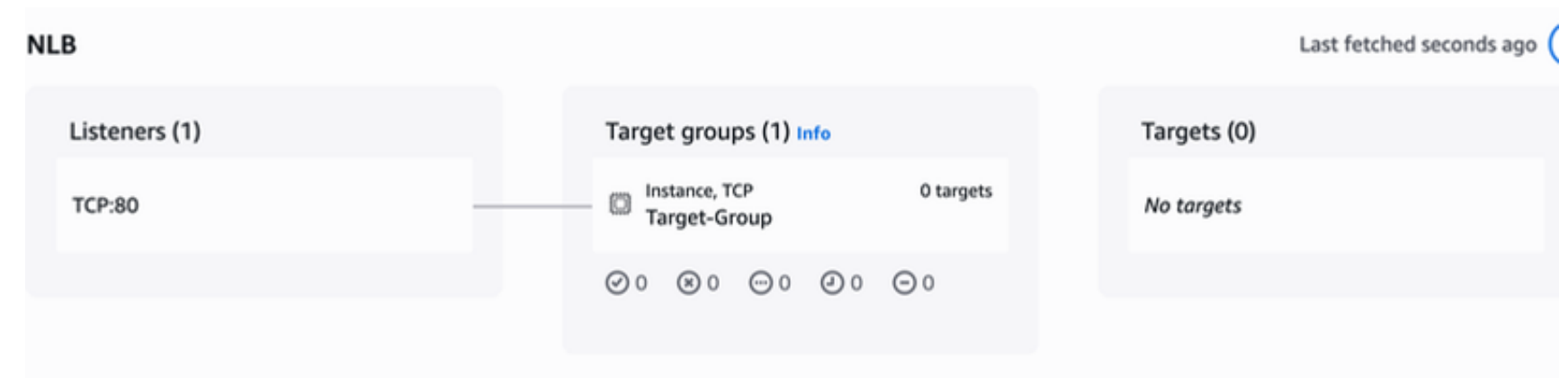
4-Tuple Used by NLB

Source IP	Source Port	Destination IP	Destination Port
203.0.113.10	51514	198.51.100.25 (NLB IP)	443

Network Load Balancer(NLB)

Benefits

- Ultra-high throughput
- Very low latency
- Preserves client source IP
- Excellent for TCP/UDP workloads



Limitations

Suppose we have:

 |company.com/api/users &  |company.com/images/logo.png

Can NLB distinguish them  **NO!**

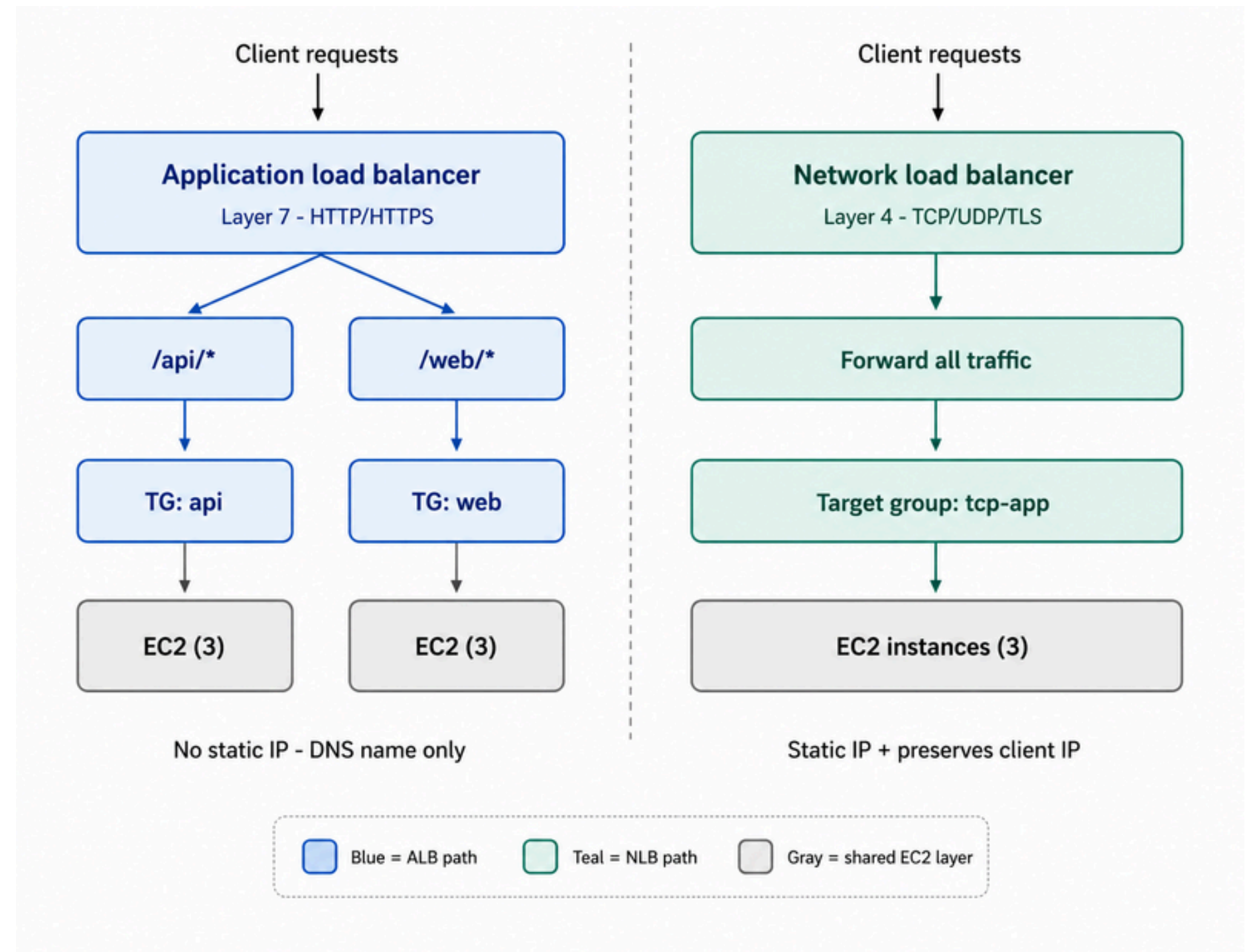
Because NLB cannot read URL

Application Load Balancer(ALB)

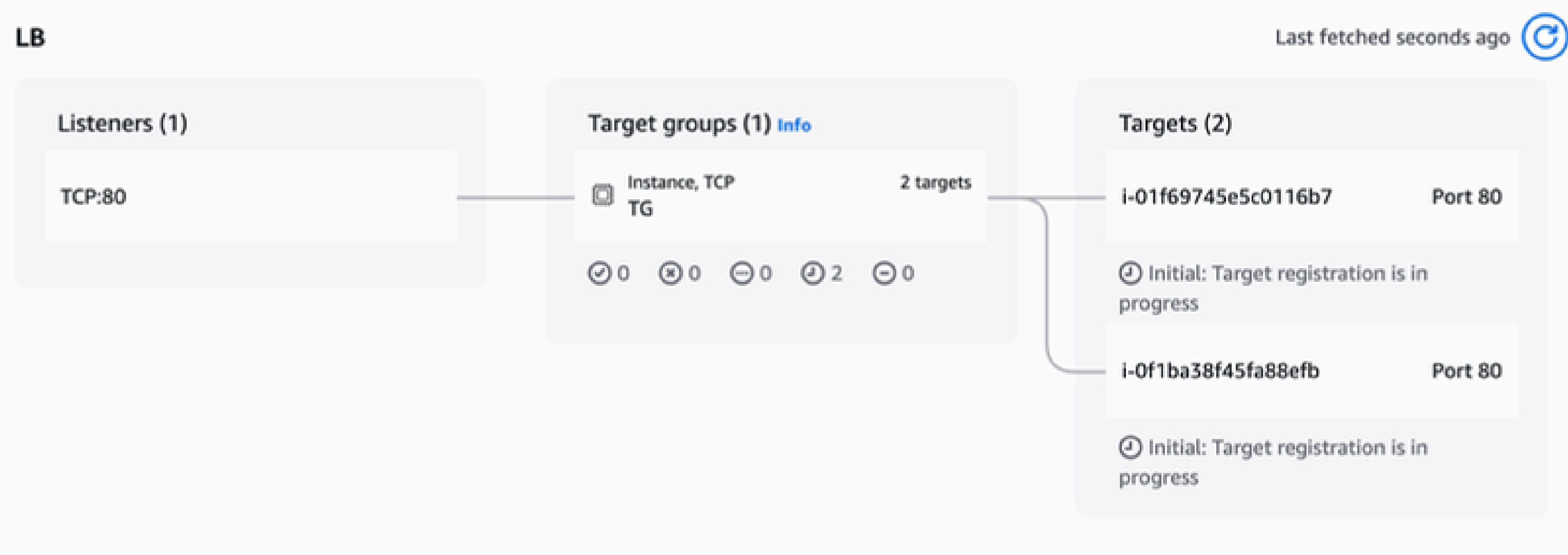
ALB = Application Load Balancer operates at Application Layer.

ALB understands

- HTTP Requests
- URLs
- Headers
- Cookies
- Host Names



Application Load Balancer(ALB)

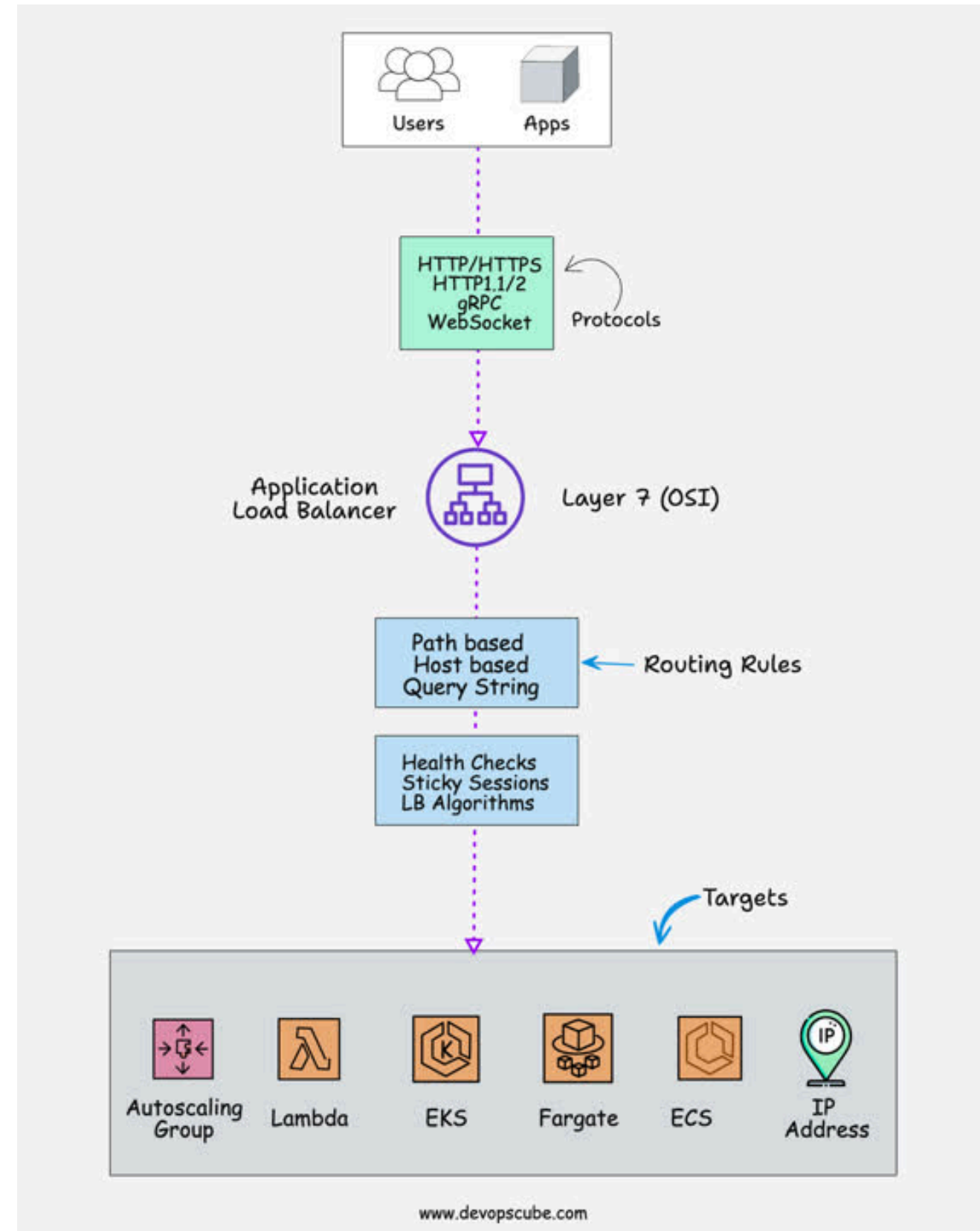
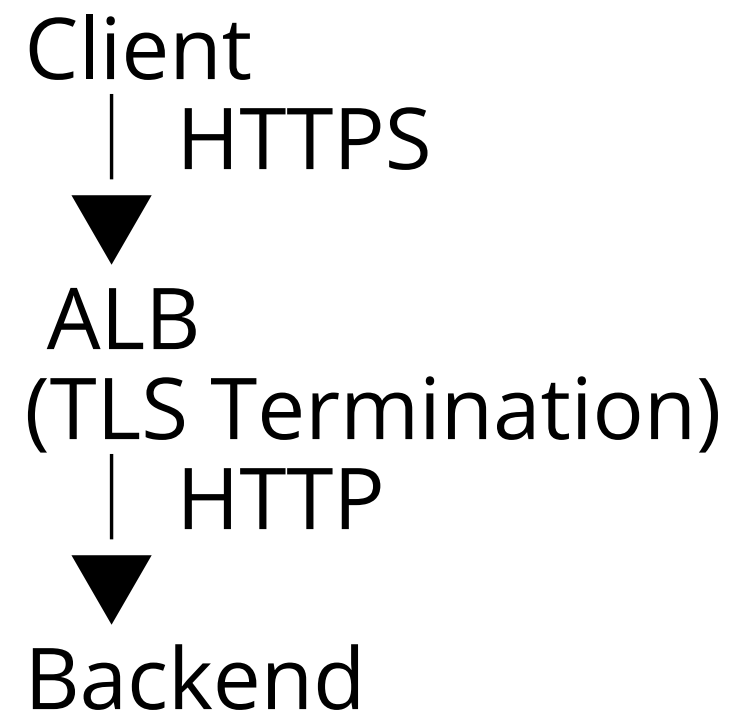


TLS Termination

HTTPS requires:

- TLS Handshake
- Encryption
- Certificates

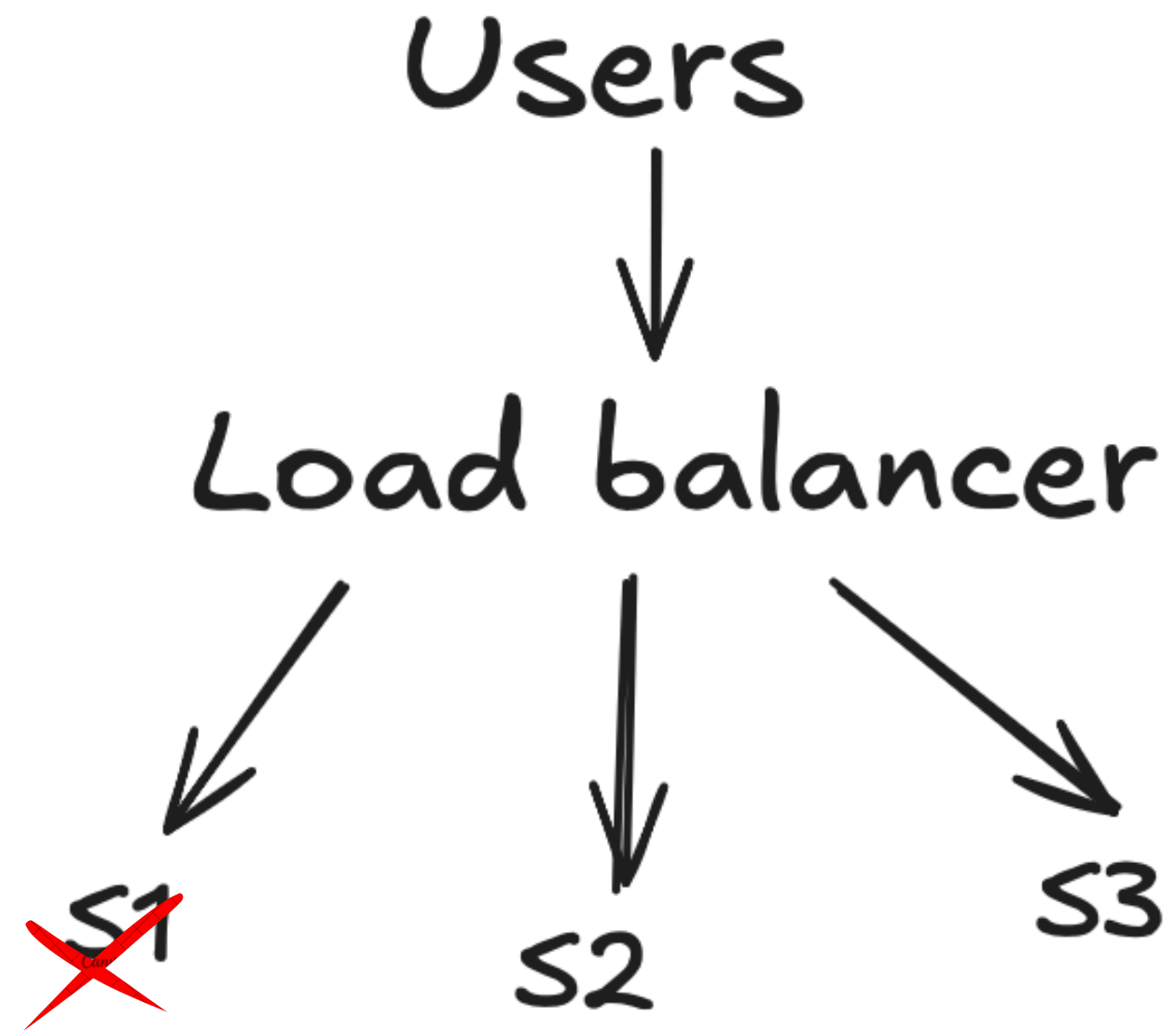
ALB can handle this.



NLB vs ALB

Feature	NLB	ALB
Layer	4	7
TCP/UDP	Yes	No
HTTP Aware	No	Yes
URL Routing	No	Yes
Header Inspection	No	Yes
TLS Termination	Limited	Yes
Source IP Preservation	Yes	Usually via headers
Latency	Lowest	Slightly higher

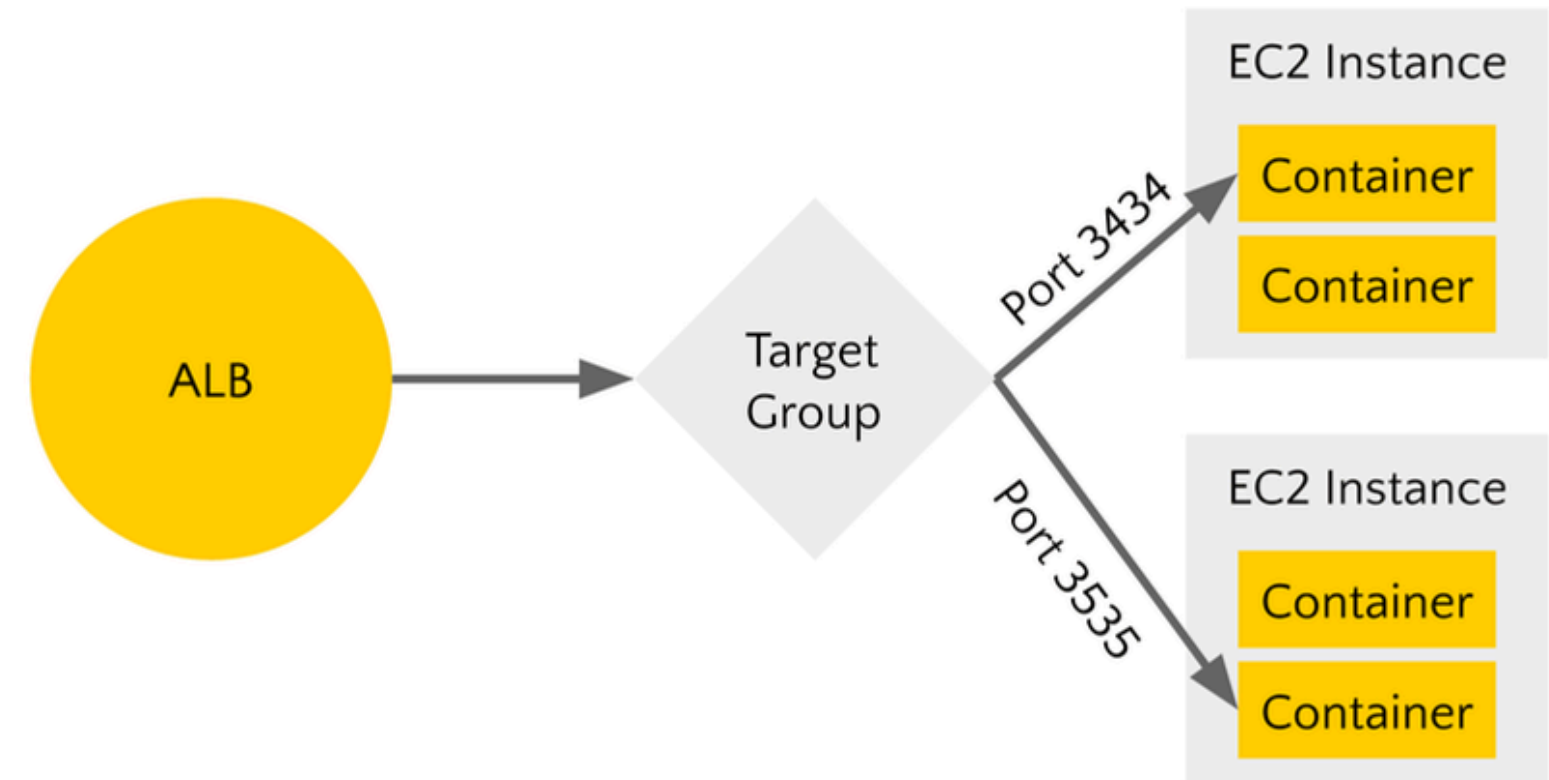
Problem



If server S1 crashes,
how does the load balancer know not to send
traffic there?

Target Group

The Target Group acts as the health-monitoring layer between the Load Balancer and the servers, allowing the Load Balancer to know which servers are healthy and which should be avoided.



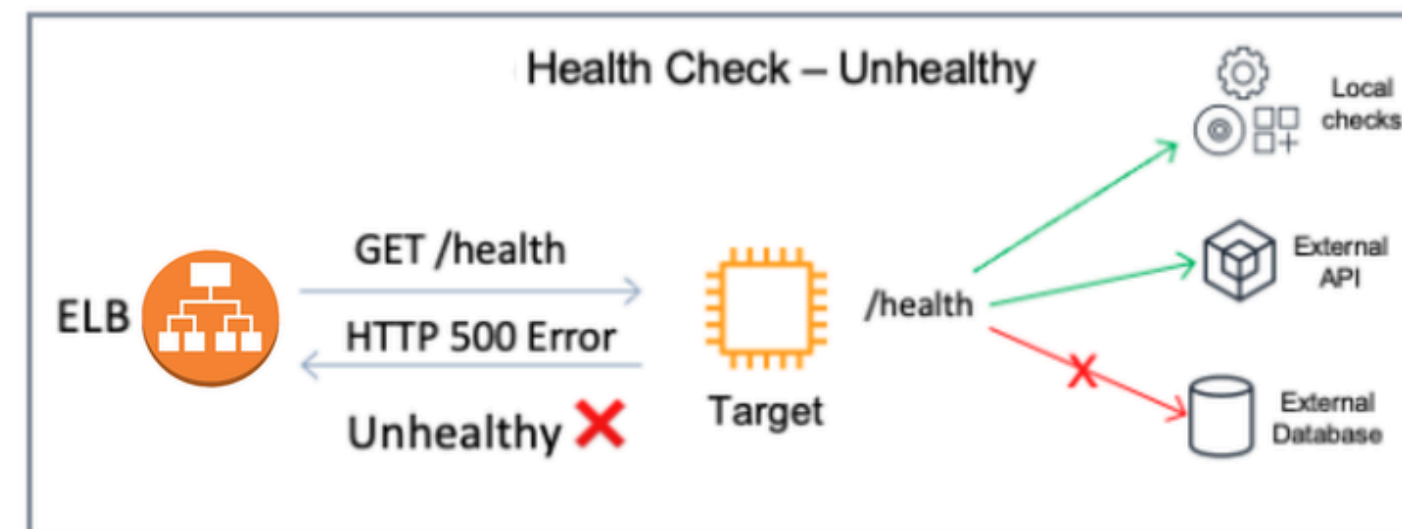
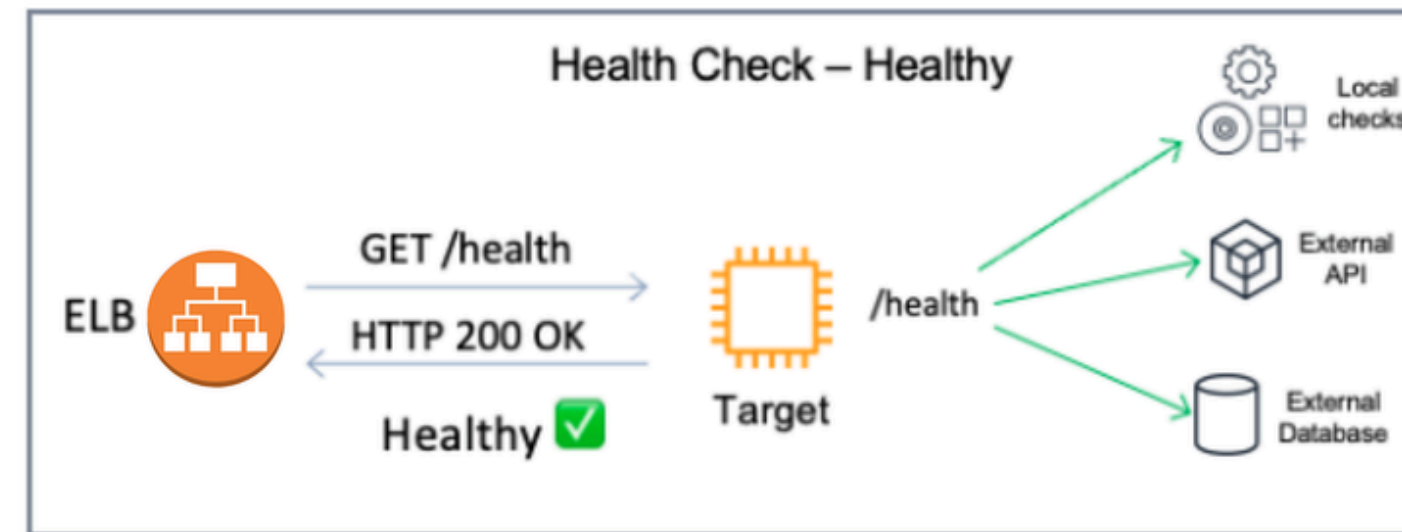
Health Checks

The Load Balancer periodically sends test requests to targets.

Example: GET /health, every: 30 sec

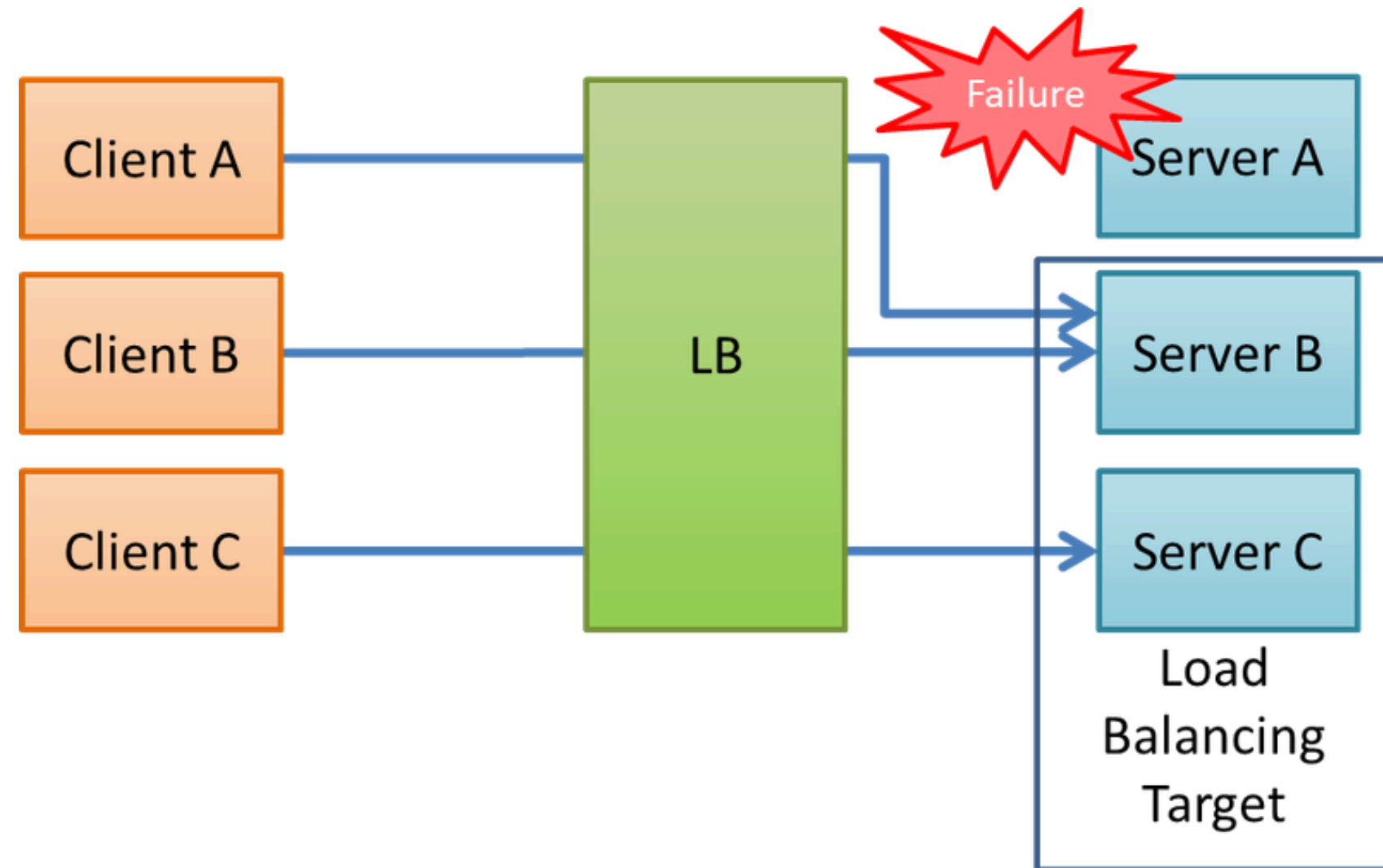
Step-by-step

1. LB sends probe
2. Server responds
3. LB marks Healthy
4. Traffic continues



Automatic Removal (Self-Healing)

Traffic no longer goes to unhealthy server.



From self healing to scalability

Health Checks



Remove Failed Servers



Self-Healing

But a new challenge appears:

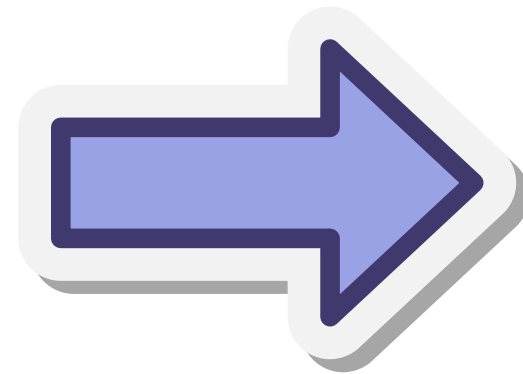
Morning Traffic = 500 Users

Evening Traffic = 50,000 Users

Question:

Should we keep 100 servers running all day?

No.



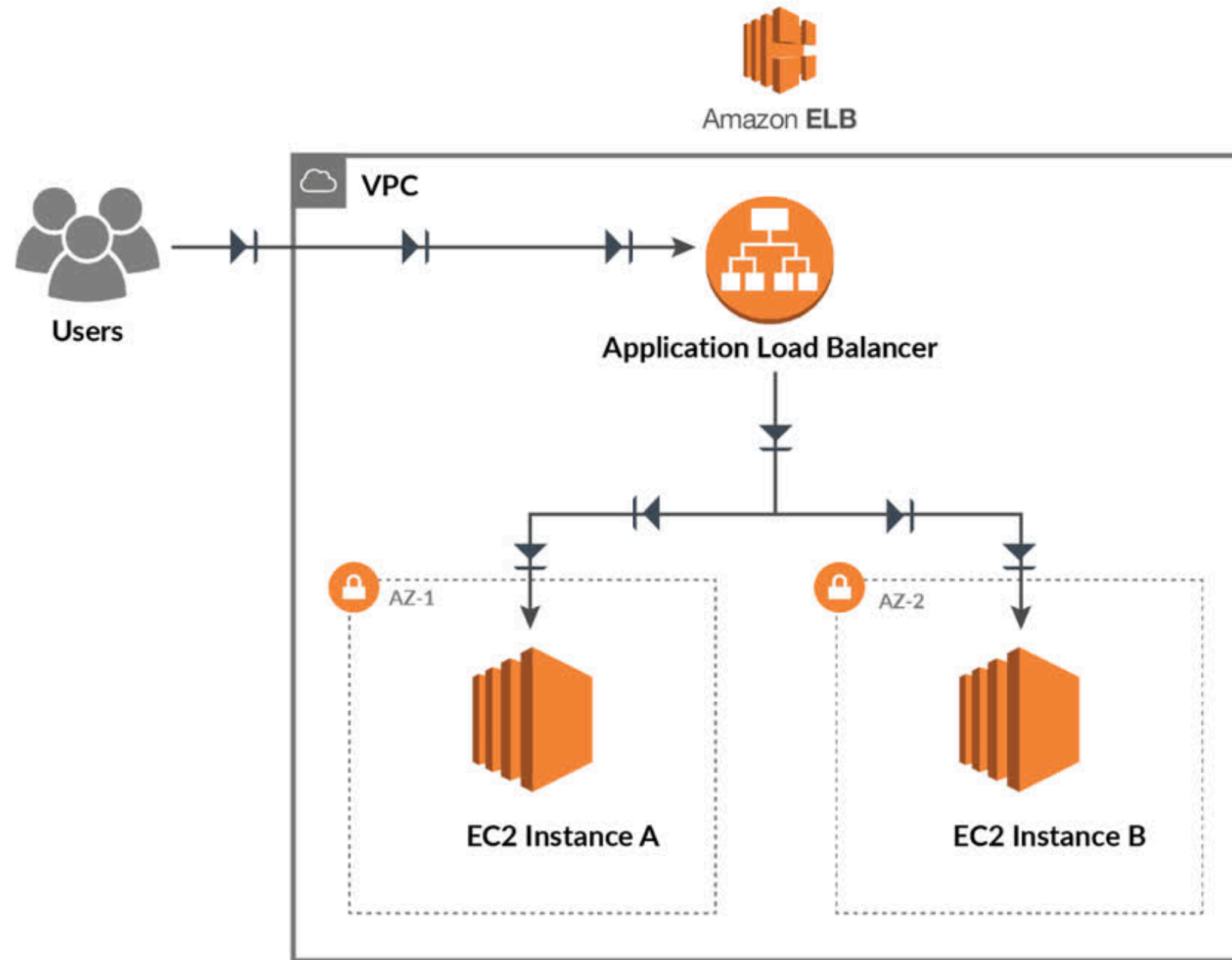
Elasticity

Elasticity means:

Increase capacity when demand rises

Decrease capacity when demand falls

Elasticity



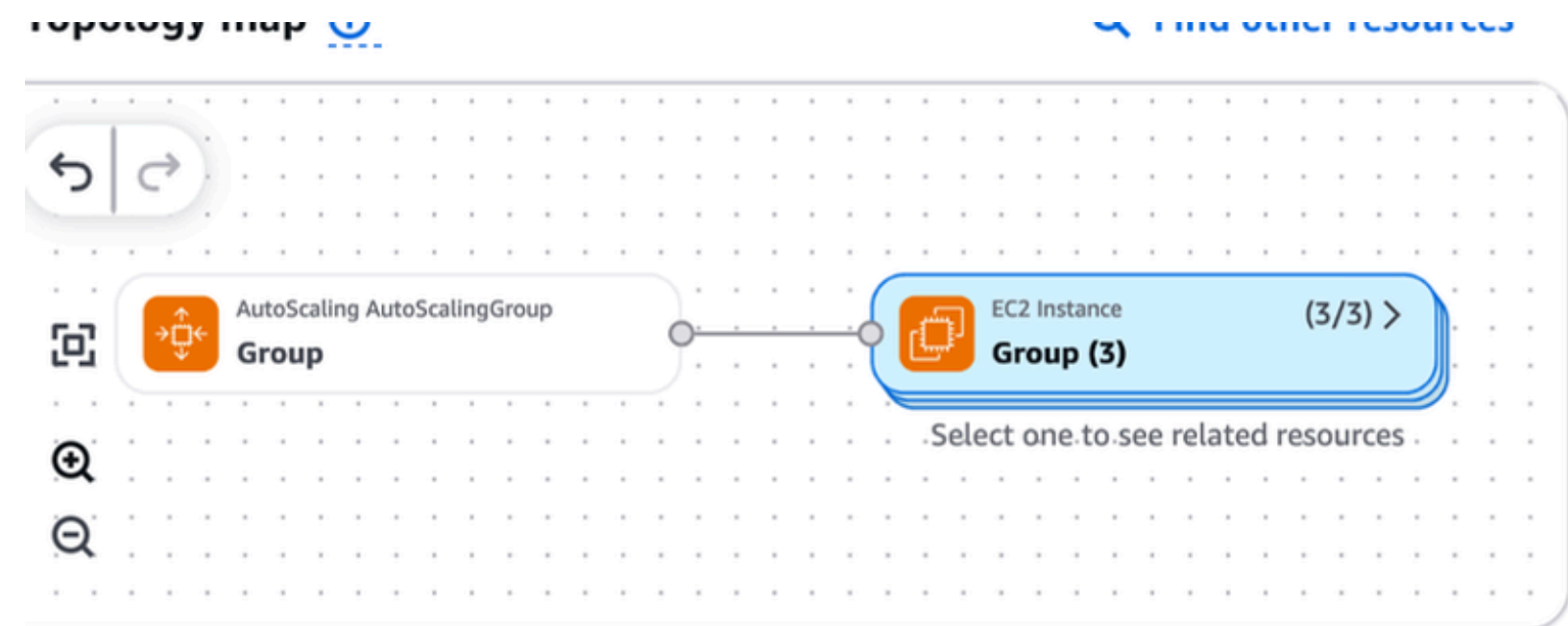
Auto Scaling Group

An Auto Scaling Group automatically:

- Launches instances
- Terminates instances
- Maintains desired capacity

When traffic increases suddenly:
Scaling Out

Before	After
EC2-1	EC2-1
	EC2-2
	EC2-3

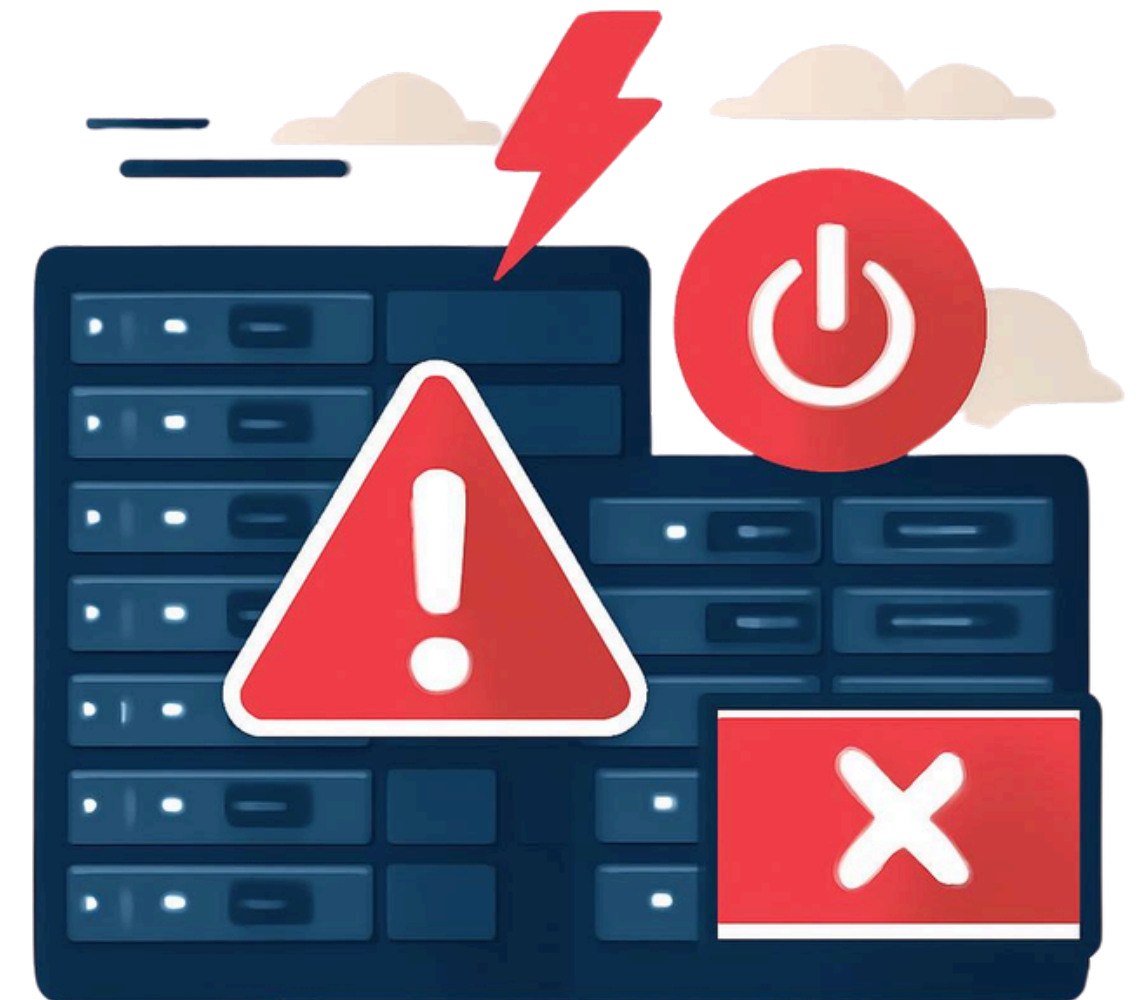


Problem — Data Center Failure

Suppose all servers are in: us-east-1a
Everything works until:

- **Power Failure**
- **Network Failure**
- **Storage Failure**

Entire application is down.



DATACENTER OUTAGE

Availability Zone

An Availability Zone (AZ) is: A physically separate AWS data center.

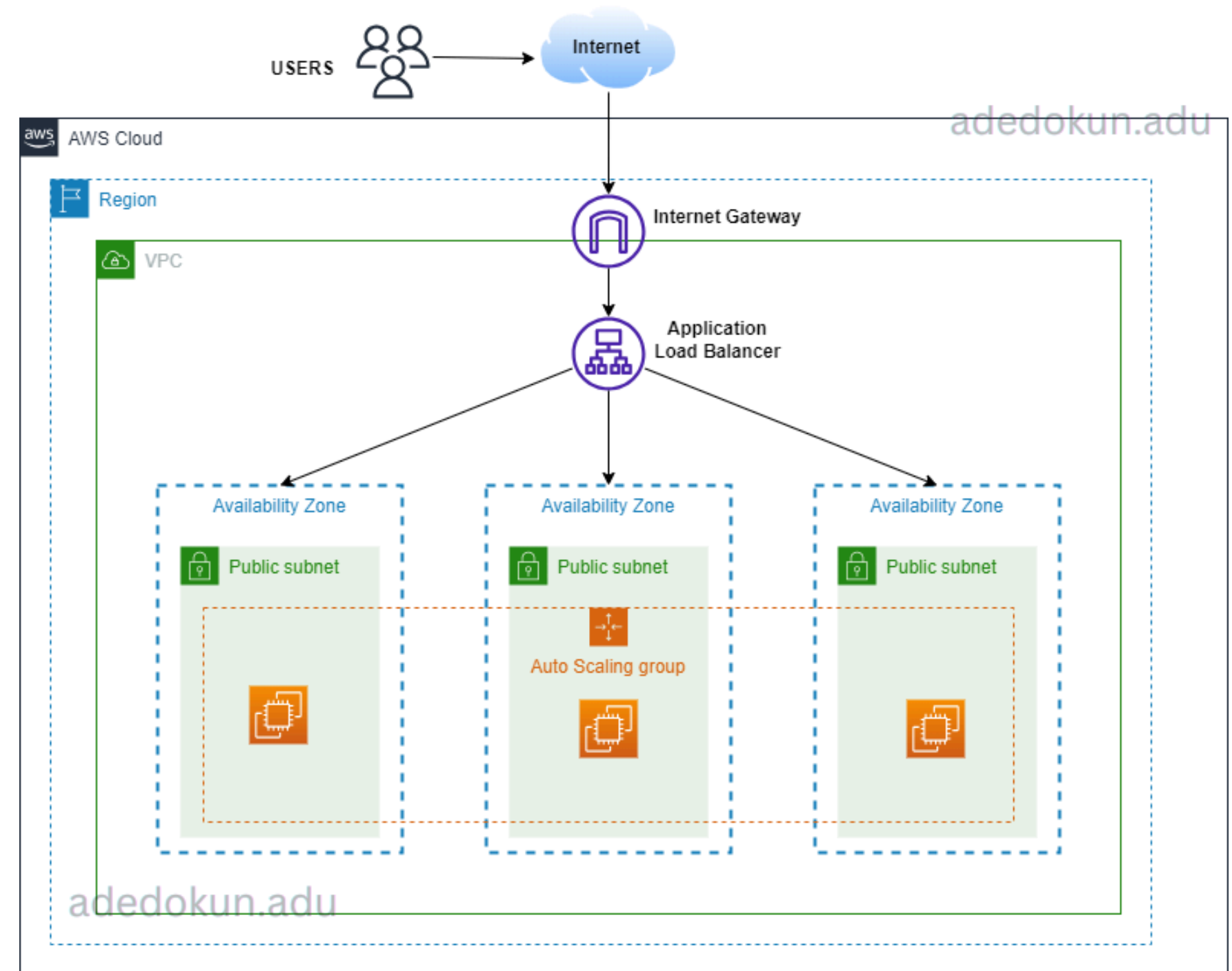
Example:

us-east-1a

us-east-1b

us-east-1c

Each AZ is isolated



Multi-Availability Zones

Availability Zones (49)

Updated 2 minutes ago

< 1 >

	Zone name	Location	Parent Region	Zone ID	Network bor...	Group name
☐	ap-southeast-6a	Asia Pacific (New Zealand)	⊖ Asia Pacific (New Zealand)	apse6-az1	–	 ap-southeast-6-z
☐	ap-southeast-6b	Asia Pacific (New Zealand)	⊖ Asia Pacific (New Zealand)	apse6-az2	–	 ap-southeast-6-z
☐	ap-southeast-6c	Asia Pacific (New Zealand)	⊖ Asia Pacific (New Zealand)	apse6-az3	–	 ap-southeast-6-z
☐	ap-southeast-7a	Asia Pacific (Thailand)	⊖ Asia Pacific (Thailand)	apse7-az1	–	 ap-southeast-7-z
☐	ap-southeast-7b	Asia Pacific (Thailand)	⊖ Asia Pacific (Thailand)	apse7-az2	–	 ap-southeast-7-z
☐	ap-southeast-7c	Asia Pacific (Thailand)	⊖ Asia Pacific (Thailand)	apse7-az3	–	 ap-southeast-7-z
☐	ap-southeast-3a	Asia Pacific (Jakarta)	⊖ Asia Pacific (Jakarta)	apse3-az1	–	 ap-southeast-3-z
☐	ap-southeast-3b	Asia Pacific (Jakarta)	⊖ Asia Pacific (Jakarta)	apse3-az2	–	 ap-southeast-3-z
☐	ap-southeast-3c	Asia Pacific (Jakarta)	⊖ Asia Pacific (Jakarta)	apse3-az3	–	 ap-southeast-3-z
☐	af-south-1a	Africa (Cape Town)	⊖ Africa (Cape Town)	afs1-az1	–	 af-south-1-zg-1
☐	af-south-1b	Africa (Cape Town)	⊖ Africa (Cape Town)	afs1-az2	–	 af-south-1-zg-1
☐	af-south-1c	Africa (Cape Town)	⊖ Africa (Cape Town)	afs1-az3	–	 af-south-1-zg-1
☐	me-central-1a	Middle East (UAE)	⊖ Middle East (UAE)	mec1-az1	–	 me-central-1-zg-

High Availability

High Availability means: Service remains available even when components fail.

Principle

No Single Point of Failure

Examples

- ✗ One server
- ✗ One AZ
- ✗ One database

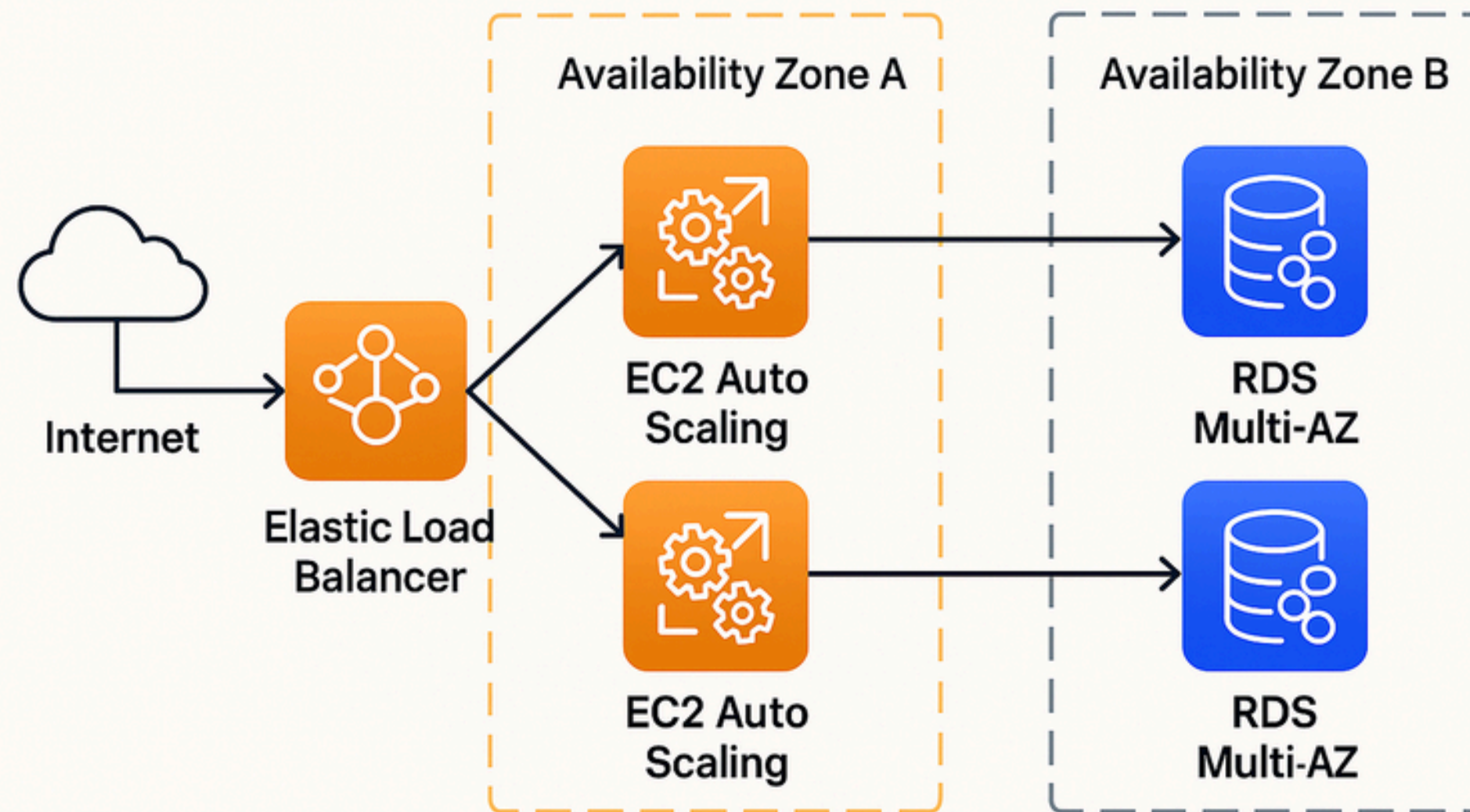
Question:

What if the load balancer itself fails?

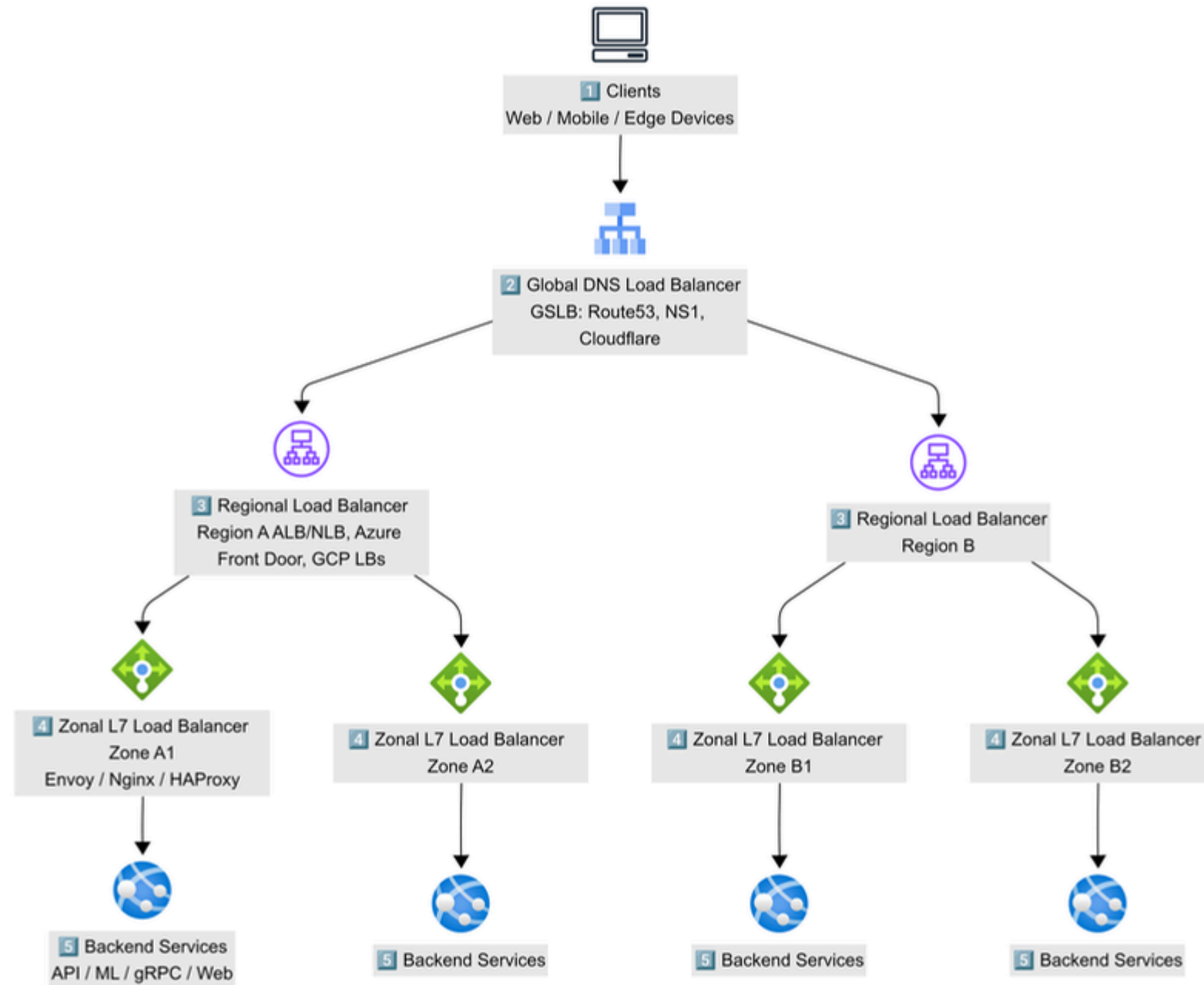
Answer:

ALB is managed by AWS
and deployed across multiple AZs

High Availability



Final Architecture Diagram



Please fill the feedback form.

Thank You